Evidence-Aware Citi Bike Forecasting and Rebalancing

Computational research project | 2025 NYC data | Reproducible analysis

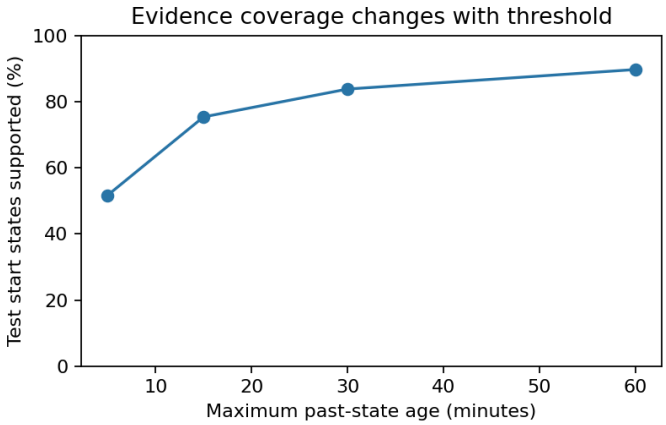
Question: can next-hour recorded-ride forecasts improve simulated rebalancing service when historical station-state evidence is incomplete and truck movements take time?

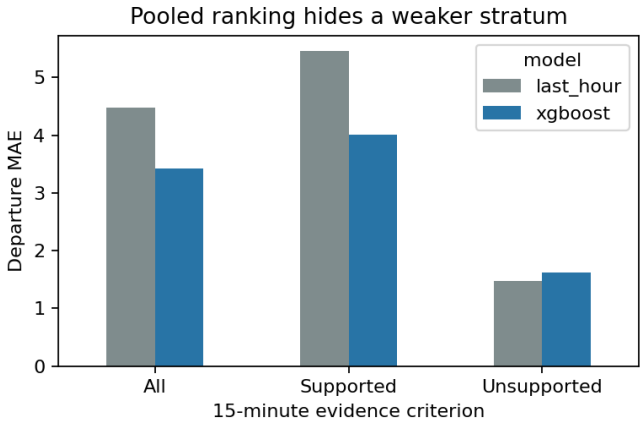
Data/method: 45.6M retained rides; 30 training-selected sites and 262,800 station-hours. Train Jan–Aug, validate Sep–Oct, test Nov–Dec; 43,950 test rows. Four fixed count models use past-only features. Metrics measure completed records, not all latent requests.

Model	Arrival MAE	Departure MAE
last_hour	4.409	4.476
seasonal_naive	5.722	5.719
poisson	4.838	4.931
xgboost	3.398	3.418

Pooled XGBoost departure MAE is 23.6% below last-hour. Historical evidence includes 2.91M state changes and dated official snapshots. The new post-hoc audit conditions on enabled, consistent source states known before the hour, with maximum age 5/15/30/60 minutes; no models are refitted or retuned.

Frozen external evaluation: protocol and fitted model hashes were committed before acquiring January 2026 records. On 22,320 new station-hours, unchanged XGBoost departure MAE is 2.705 versus 3.235 for last-hour (16.4% lower). All outcomes are retained; no refitting or tuning occurs.





At 15 minutes, 33,131 starts are supported and 10,819 are unsupported. On the latter, XGBoost departure MAE 1.621 exceeds last-hour 1.480. Conditional performance is heterogeneous; differences in activity and evidence sampling preclude a causal explanation. Start-state support does not verify continuous operation through the future hour.

Decision models, validation and limits

Methods: constrained integer rebalancing, three-period/four-scenario stochastic MPC, shared-request count simulation and paired recorded-ride replay. The full paired test spans 1,465 hours and handles rejected departures, cancelled returns, transit, external boundaries and held bikes.

Delayed road experiment: 48 hours from the first eligible common historical anchor; two capacity-six co-loaded pickup-delivery trucks, directed OSRM driving times, 90 s/bike loading/unloading, hourly depot returns and driver/budget limits. Cargo changes inventory only at modeled pickup/delivery times. Independent checks reconcile 181 routes and actual cargo ledgers.

Policy	Unserved records	Truck km	Service proxy %
none	1350	0.0	94.776
road_greedy	1442	1068.85	94.42
road_milp	1422	1171.18	94.498

Finding: road MILP does not beat no intervention in this scenario. Optimal forecast-end balance under a proxy objective does not imply improved realized service. Current driving times, assumed depot, frozen usable capacity and blocked-return rules are scenario assumptions; this is not a measured operator effect.

Unresolved evidence: 19,632 station-hours (7.47% of the annual grid) do not meet the strong historical sampling criterion. This is a different criterion/time scope from the 10,819 unsupported test starts. Change-filtered archives suppress unchanged-count and possibly flag-only updates. Missing observations remain unknown. Complete inventories would still not reveal every rejected request.

Inference limits: the 2025 age audit is post-hoc; the separate January 2026 evaluation freezes protocol before data acquisition. Both use archived-count rolling one-step inputs, not a deployed online forecast. Comparisons are fair within the same rows/scenario. Different strata, dates or capacities are not interchangeable treatment groups. Daily bootstrap intervals are conditional, not causal guarantees.

Engineering: 60 local tests pass; public Research checks CI succeeded. Prospective GBFS collection stores unchanged counts, flag-only changes and failure heartbeats, with verified archives/restoration. Scheduling and cadence are audited from actual receipts; no long-term uptime claim is made.

Reproduce: install exact requirements-lock.txt; run python -m pytest. Full research commands and data manifests are in README. scripts/missingness_experiment.py regenerates this audit; scripts/check_transport.py validates saved road results. The portable delivery contains the road window inputs and supports --reuse-rides.

Repository: https://github.com/ryanwulikenature-glitch/ryan-wu

Sources: Citi Bike System Data; CityBikes historical archives (ODbL); official GBFS captures via Internet Archive; OpenStreetMap/OSRM driving proxy. Full attribution and empirical claim sources accompany the repository.

Contributions and tool use | Computational research project; empirical scope and scenario assumptions are described above.